🧠 Entrepreneurship Bootcamp — Complete Study Guide

Cheatsheet by Makram ElJamal for Entrepreneurship Bootcamp · Spring 2026

Each card is tagged with the Pearson criteria it covers: **P#** Pass · **M#** Merit · **D#** Distinction. Colour of card: **teal**=business · **rose**=design · **amber**=formula/exam-critical · **indigo**=worked example.

📊 BUSINESS TRACK — Zaid Maaytah · Lectures 1–9

L1 · Innovation & Entrepreneurship P1

Innovation = creating **and** implementing new ideas that generate value (economic, social, or technological). Can be a new product, service, process, or business model.

Key idea: innovation ≠ invention. An invention is a new thing; an innovation is a new thing that **creates value** and is actually used. No value = not innovation.

Entrepreneurship = identifying opportunities, mobilizing resources, and creating value through new ventures. It turns ideas into sustainable businesses, and is the **4th factor of production** alongside land, labor, capital — the function that organizes the other three.

Innovation Sweet Spot — good innovation sits where 3 circles overlap:

- **Desirability** — do people actually want it? (human lens)
- **Feasibility** — can it technically be built/delivered?
- **Viability** — can the business make money & sustain itself?

⚡ Missing one: desirable+feasible but not viable = no business; desirable+viable but not feasible = can't build it.

L1 · Design Thinking Process & DT↔Business P3 P4

Design Thinking (Stanford d.school) — a 5-stage process that starts with **people and their needs**, not tech or a business plan:

Empathize → Define → Ideate → Prototype → Test

- **Empathize** — understand the user deeply
- **Define** — frame the real problem
- **Ideate** — generate many possible solutions
- **Prototype** — build cheap, rough versions
- **Test** — put them in front of users and learn

Airbnb: bookings were flat → listing photos were poor → sent photographers → bookings rose. A **user** fix, not a tech fix.

⚡ Design Thinking decides **WHAT** to build; Entrepreneurship decides **HOW** to deliver & scale it.

L1 · 3Ms, Mindset, Agility & Teams P1

The 3Ms — what every founder manages together:

- **Mind** — mindset; sets direction, resilience, adaptability
- **Market** — the map; where value is & who needs it
- **Money** — the fuel; manage it wisely or you stall

Growth Mindset = ability grows through effort; failure = **learning**. Traits: Curiosity, Vision, Risk Tolerance, Resilience, Adaptability, Persistence.

Business Agility = adapt quickly to market/tech/customer change via rapid experimentation, continuous learning, iterative development, flexible decisions.

High-Performance Teams collaborate, not just have talent. **CATme** = peer-feedback system → raises accountability, exposes contribution imbalances.

L2 · PEST Analysis M1

PEST = scan of the external **macro-environment** — big forces outside the company you can't control but must respond to. Used when evaluating a new market.

Political	Policy, regulation, stability, support for entrepreneurship, licenses, public funding
Economic	Inflation, interest rates, (un)employment, purchasing power, capital/credit access
Social	Cultural values, demographics, education/literacy, lifestyle & social trends
Technological	Internet/mobile, infrastructure, emerging tech, digital tools/platforms

⚡ PEST ≠ PESTEL (which adds Environmental + Legal). PEST feeds the **Opportunities & Threats** of a SWOT.

L2 · SWOT Analysis M1

	Positive	Negative
Internal	**Strengths** — what you do well	**Weaknesses** — what you lack
External	**Opportunities** — trends/gaps	**Threats** — competition, regulation

How to classify: (1) is it **inside** the company or **outside**? (2) good or bad? New competitor cutting prices = outside + bad = **Threat**. Strong team = inside + good = **Strength**.

⚡ S+W = internal; O+T = external. PEST informs the O and T.

L3 · Entrepreneurship Approaches P2 D1

Lean Startup — kill waste by testing before scaling. **Build** → **Measure** → **Learn**; build an **MVP** (smallest thing that tests your key hypothesis), measure real behaviour, then **pivot** or **persevere**. *Dropbox tested demand with a demo video before coding.*

Agile — iterative **sprints**, continuous delivery + feedback, flexible direction.

Tech-led vs Design-led — "built it because we could" vs "because they needed it." Best = both, but start from human need.

Social Entrepreneurship — social/environmental value via business methods. *Grameen Bank microfinance.*

⚡ Lean & Agile **complement** each other. Limited resources + uncertain market + likely pivot → choose **Lean** (D1: justify by integrating O&T, resources, team, goals).

L4 · Business Model Canvas — 9 Blocks P8 D4

BMC = one-page tool to visualise how a business creates, delivers & captures value. Right = market, centre = value, left = operations, bottom = money. Asks: Desirable? Feasible? Viable?

Customer Segments	Who you serve (demo/geo/behaviour/psycho). No segment = no customer.
Value Proposition	Unique value; what problem & why different. Ties to Jobs/Pains/Gains.
Channels	Reach/deliver: sales, communication, distribution.
Customer Relationships	Personal / self-service / automated.
Revenue Streams	Sales, service fees, subscription, licensing, advertising.
Key Resources	Physical, intellectual (IP), human, financial.
Key Activities	Production, marketing/sales, distribution, service.
Key Partners	Suppliers, manufacturers, distributors, alliances, investors.
Cost Structure	Fixed + variable; economies of scale & scope.

⚡ Mistakes: too many segments (pick a **beachhead**), vague VP, unrealistic revenue, ignoring costs. **D4 "best single change"** = one fix that improves desirability + feasibility + viability at once (e.g. a key partner that fixes the core failure & unlocks recurring revenue).

L5 · Value Proposition Canvas P6

Zooms into Value Proposition ↔ Customer Segment to check they match.

Customer Profile (right): Jobs (functional/emotional/social) · **Pains** (frustrations, risks) · **Gains** (desired outcomes).

Value Map (left): Products & Services · **Pain Relievers** · **Gain Creators**.

Fit = Pain Relievers address real Pains and Gain Creators deliver real Gains.

VP statement: "[Product] helps [customer] achieve [benefit] by [solving pain]." *Uber: a fast, reliable, affordable ride at the tap of a button — no waiting, no negotiation, no cash.*

⚡ "Saves an hour a day" = a **Gain**. "Reduces long delivery waits" = relieving a **Pain**.

L5 · Markets & B2B vs B2C P5

Market = system where buyers & sellers exchange. Types: consumer, financial, business, online.

B2B (to businesses)	B2C (to consumers)
Rational, ROI-driven	Emotional, personal
Long cycles	Short cycles
Many decision-makers	Usually one buyer

⚡ If the **user isn't the payer** you're in a "system game" with B2B dynamics — design for the decision-maker.

L5 · Diffusion of Innovation P5

Innovators 2.5%	Risk-takers; try first, before it's proven
Early Adopters 13.5%	Visionaries/opinion leaders; spot benefit early
Early Majority 34%	Pragmatists; risk-averse; adopt once proven
Late Majority 34%	Skeptics; price-sensitive; adopt late
Laggards 16%	Traditionalists; last to adopt

⚡ Your **SOM** (first customers) = Innovators + Early Adopters (~16%) — they tolerate an unfinished product.

L5 · ★ TAM / SAM / SOM — Market Sizing P5

TAM = max demand if 100% adopt (whole pie). **SAM** = slice you can realistically serve (geo/constraints). **SOM** = slice you actually capture short-term (1–3 yrs), your "entry wedge."

$\text{SAM} = \text{TAM} \times \text{target-segment}\,\%$ $\text{SOM} = \text{SAM} \times \text{market-share (conversion)}\,\%$

Worked example

3,000,000 smartphone users; 60% health-conscious; 25% willing & able to pay; 10% Year-1 conversion.

- TAM = **3,000,000**
- SAM = 3,000,000 × 60% × 25% = **450,000**
- SOM = 450,000 × 10% = **45,000**
- Annual SOM revenue = users × monthly price × 12

⚡ Higher price → fewer "willing & able" → **SAM shrinks → SOM shrinks**. Cap SOM at operational capacity.

L6 · Competitor Analysis & USP P6

Competitor types: Direct (same product+audience), **Indirect** (different product, same need), **Emerging** (disruptive new entrants).

Benchmarking = comparing vs rivals. A **Moon Chart** (radar/spider chart) plots several competitors across several attributes at once.

Perceptual / Positioning Map = plot brands on 2 attributes (e.g. Price vs Quality); empty regions = **market gaps** = opportunities.

USP = the standout feature/benefit. Must be **Unique, Valuable, Clear** — where "what customers want" meets "what rivals do poorly."

⚡ Benchmarking reveals where you outperform → that gap becomes your USP.

L6 · ★ Pricing Strategies M3

Cost-Plus	**Price = cost + markup %.** Simple, guarantees a margin; ignores willingness to pay.
Value-Based	Price on **perceived value**, not cost. Needs strong differentiation. Luxury/branded.
Competitive	Price vs rivals (below/at/above). Used when products are similar.
Penetration	**Low** entry price to grab share fast, raise later. Price-sensitive markets. Risk: price-chasers.
Skimming	**High** price first (premium early adopters), lower over time. *New iPhone.*
Psychological	Feels cheaper/better: charm ($9.99), prestige, bundling, anchoring, scarcity.

⚡ Risk-averse, budget-conscious buyers at launch → **penetration + strong promotion**; premium/skimming would be rejected.

L7 · 4Ps of Marketing (→ 7Ps) M3

The **Marketing Mix** — levers to take a product to market; each P must align with the others & the target customer.

Product	Features, quality, design, branding. Core (benefit) vs augmented (warranty). Lifecycle: Intro→Growth→Maturity→Decline.
Price	What customers pay; signals value, positions brand.
Place	How & **where/when** they access it: direct / indirect / digital. *New delivery windows/routes = Place.*
Promotion	Communicate value: advertising, PR, social, content, sales promos. IMC = one consistent message.

7Ps (services) add **People**, **Process**, **Physical Evidence**.

⚡ Classify: discount = **Price**; meal bundle = **Product**; influencer = **Promotion**; evening delivery = **Place**.

L8 · Cost, Revenue & Finance Basics P7

Entrepreneurship = creativity + **resource management**. Finance helps you survive, plan, grow, reduce uncertainty. Most startups fail from **financial mismanagement**, not bad ideas.

Common mistakes: underestimating costs, overestimating revenue, spending before validating, poor cash-flow management.

Startup Costs	One-time, before revenue: equipment, legal, website, inventory, licenses.
Fixed Costs	Don't change with volume: rent, salaries, insurance, software.
Variable Costs	Scale with output: materials, packaging, delivery, commissions.

$\text{Total Cost} = \text{Fixed} + (\text{Variable/unit} \times \text{Units})$

Projected Revenue answers WHO buys, WHAT, at what PRICE. Use a **bottom-up** estimate; avoid the "just get 1% of the market" fallacy.

💰 FINANCE DEEP-DIVE — L9 + the Merit/Distinction calculation engine

The Three Financial Statements M4

Income (P&L)	Revenue − Costs = Profit/Loss **over a period**. Shows profitability.
Balance Sheet	**Assets = Liabilities + Equity** at a single **point in time**.
Cash Flow	Actual cash **in & out** over a period — not the same as profit.

Key terms: Revenue = total sales income. **COGS** = direct cost of what you sold. **OpEx** = running costs (marketing, admin, rent).

⚡ Profitable ≠ cash-positive: you can book a sale before the customer pays. **Burn rate** = how fast reserves are spent.

Income Statement (P&L) — layout M4

A "waterfall": from sales, subtract costs in layers down to the bottom line. Covers a **period**.

Revenue (Sales)	1,000,000
− Cost of Goods Sold (COGS)	(600,000)
= Gross Profit	**400,000**
− Operating Expenses (OpEx)	(250,000)
= Operating Profit (EBIT)	**150,000**
− Interest	(20,000)
− Tax	(30,000)
= Net Profit	**100,000**

Tips: work **top-down** (each subtotal feeds the next). Only **direct production** costs in COGS; running costs (rent, marketing, admin, depreciation) in OpEx.

⚡ GPM = 400,000/1,000,000 = 40%; NPM = 100,000/1,000,000 = 10%.

Balance Sheet — layout M4

A **snapshot** at one date. Always balances: **Assets = Liabilities + Equity**.

ASSETS	
Cash	50,000
Accounts receivable	70,000
Inventory	80,000
Current assets	**200,000**
Equipment (non-current)	300,000
Total Assets	**500,000**
LIABILITIES	
Accounts payable	90,000
Short-term loan	60,000
Current liabilities	**150,000**
Long-term loan	150,000
Total Liabilities	**300,000**
EQUITY	200,000
Liabilities + Equity	**500,000**

Tips: solve a missing figure via **Equity = Assets − Liabilities**. **Current** = within 1 yr; **non-current** = long-term. Current Ratio = 200,000/150,000 = **1.33**.

⚡ If it doesn't balance, you missed or mis-signed an item — recheck.

Cash Flow Statement — layout M4

Actual **cash in/out** over a period, in three buckets, ending at the bank balance.

OPERATING (trading)	
Cash from customers − cash paid	120,000
INVESTING (assets)	
Bought equipment	(300,000)
FINANCING (loans/equity)	
Loan + investor funds	250,000
Net change in cash	**70,000**
+ Opening cash	30,000
= Closing cash	**100,000**

Tips: sort each item into **Operating** (sales, salaries, rent), **Investing** (equipment, property), or **Financing** (loans, equity). **Closing = Opening + Net change**; outflows are negative.

⚡ A profitable firm can be cash-negative (big purchase / unpaid invoices) — only this statement reveals it.

★ Core Formulas (with meaning) D3

$$\text{Gross Profit} = \text{Revenue} - \text{COGS}$$

Left after the direct cost of making the product — production efficiency.

$$\text{Net Profit} = \text{Gross Profit} - \text{OpEx} - \text{interest} - \text{tax}$$

The true bottom line after **all** costs.

$$\text{Gross Profit Margin} = \text{Gross Profit} \div \text{Revenue} \times 100$$

$$\text{Net Profit Margin (NPM)} = \text{Net Profit} \div \text{Revenue} \times 100$$

Profit as a **% of revenue** — compares companies of any size.

$$\text{ROI} = (\text{Gain} - \text{Cost}) \div \text{Cost} \times 100 = \text{Net Profit} \div \text{Invested} \times 100$$

Return per dinar invested. Higher = more efficient.

$$\text{Current Ratio} = \text{Current Assets} \div \text{Current Liabilities}$$

Liquidity — can you cover short-term debts? ≥1.5 = common threshold.

★ Breakeven Analysis P7 D3

The point where **Total Revenue = Total Costs** (zero profit). "How much must we sell to stop losing money?"

$$\text{Contribution Margin (CM)} = \text{Price} - \text{Variable/unit}$$

$$\text{Breakeven Units} = \text{Fixed Costs} \div \text{CM}$$

$$\text{Breakeven Revenue} = \text{Breakeven Units} \times \text{Price}$$

Worked — profit/loss position

Café: Fixed 40,000 · Price 4 · Variable 1.50 · Sold 12,000.

- CM = 4 − 1.50 = **2.50/cup**
- Total contribution = 12,000 × 2.50 = **30,000**
- Profit = 30,000 − 40,000 = **−10,000 (loss)**
- Breakeven = 40,000 ÷ 2.50 = **16,000 cups**

⚡ Trap: compare **total contribution** (CM×units) to fixed costs — not revenue to fixed costs.

Worked: Net Profit with adjustments M4 D3

Sales 1,000,000 · COGS 520,000 · OpEx 280,000. Adjust: −15,000 sales; move 30,000 delivery COGS→OpEx; +20,000 depreciation to OpEx.

- Sales = 1,000,000 − 15,000 = **985,000**
- COGS = 520,000 − 30,000 = **490,000**
- OpEx = 280,000 + 30,000 + 20,000 = **330,000**
- Gross Profit = 985,000 − 490,000 = **495,000**
- Net Profit = 495,000 − 330,000 = **165,000**

⚡ Order: adjust Sales → COGS → Gross Profit → OpEx → Net Profit.

Worked: Reclassification & margins D3

Move a cost (e.g. shipping) **COGS → OpEx**, nothing else changes:

- COGS falls → Gross Profit rises → **GPM increases**
- Total expenses unchanged → **Net Profit & NPM stay the same**

Reverse (OpEx→COGS): GPM **down**, NPM same. Add **depreciation to OpEx**: NPM **down**, GPM unchanged.

⚡ Golden rule: **moving** a cost between lines never changes Net Profit — only **adding/removing** one does.

Worked: Price rise vs cost cut D3

Price 50 · unit COGS 32 · fixed OpEx 120,000 · volume 10,000. Opt 1: price→52. Opt 2: COGS→30.

- **Opt 1:** Rev 520,000; NP = (52−32)×10,000 − 120,000 = 80,000; NPM = **15.4%**
- **Opt 2:** Rev 500,000; NP = (50−30)×10,000 − 120,000 = 80,000; NPM = **16.0%**

⚡ Same Net Profit on **lower revenue** = higher margin → the **cost cut wins**.

Worked: Viability thresholds D3

Lender wants NPM ≥ 12%; investor wants ROI ≥ 20%. Revenue 2,400,000 · COGS 1,380,000 · OpEx 960,000 · avg invested 150,000.

- Net Profit = 2,400,000 − 1,380,000 − 960,000 = **60,000**
- NPM = 60,000 / 2,400,000 = **2.5% → FAILS 12%**
- ROI = 60,000 / 150,000 = **40% → PASSES 20%**

⚡ NPM uses revenue, ROI uses invested capital — test each threshold separately.

Worked: ROI comparison D3

Company A: invested 80,000, net profit 12,800. Company B: invested 60,000, net profit 9,000.

- A: 12,800 / 80,000 = **16%**
- B: 9,000 / 60,000 = **15%**
- **A wins** — higher return per dinar.

Bigger absolute profit ≠ better — divide by invested (ROI) or revenue (margin) to compare fairly.

Worked: Comparing two revenue options D3

Pick the option with the higher NPM. For each option:

- Sum all **new revenue** (base + each new stream)
- Sum all **variable costs** (per-unit × units)
- Sum all **fixed** costs (base + extra support)
- NP = Rev − Variable − Fixed; NPM = NP ÷ Rev

"Additive" = stack the new stream on the baseline, recompute the whole P&L, then compare margins. A high-price/low-cost subscription usually beats a low-fee stream.

DESIGN THINKING TRACK — Fayzeh Abuhaltam · Lectures 1–7

L1 · Why Startups Fail & HCD P3

Why startups fail: build for themselves not users; start with a solution not a problem; same-background teams share biases & blind spots.

Bad: Idea → Build → Hope. **Good: People → Problem → Solution.**

Human-Centered Design = start with people — frustrations, needs, behaviour, habits, workarounds. "*Users describe solutions; designers discover needs.*"

DT = "**observation without judgment**." Your own expertise is a filter that can stop you seeing reality.

L1 · Cognitive Biases P3

Halo Effect	One impressive feature hides the flaws
Similarity (Like-Me)	Assuming users behave like you
Confirmation	Seeking only info that supports your idea
Stereotyping	Over-generalising a user group

These make you "validate" assumptions instead of genuinely testing them.

L1 · Human Hacks & Design Journey P3

Human Hack = a **workaround** people create because a system failed a need (infra/design/trust/process). **Where there's a workaround, there's an unmet need.** *e.g. propping a laptop on folded paper.*

Understand → Research → Define → Ideate → Prototype → Test → Evaluate → Pitch

"*Business is the destination; Design is the maze — don't jump over it, walk through.*"

Decode complaints deeper: "slides earlier" → need for **control**; "hate group projects" → **fair workload**; "bigger battery" → **anxiety reduction**.

L2 · Observe · Engage · Immerse P3

Self-reporting fails: people don't do what they *say*, *think*, or are *told*.

Observe	Watch in context — habits, workarounds, friction, hesitation. Don't intervene.
Engage	Ask **behavioural** questions about real past events, not opinions. Decode signals (phone = boredom).
Immerse	Experience it yourself → how it **feels**. *Write with your non-dominant hand.*

Observed behaviour > stated preference. Observation = *what*, talking = *why*, immersion = *how it feels*.

L2 · Primary vs Secondary Research P3

Primary (direct): first-hand — observe in context, behavioural interviews, immersion.

Secondary (indirect): existing data — reports, statistics, articles. Replaces assumptions with evidence.

Every insight should answer: **who** is the user, **what** do they need, in **what context**?

"*You think you understand a problem — but you only saw a symptom.*"

L3 · PESTEL & System Thinking M1

Business sees the **MAP** (where to go); Design sees the **TRAP** (what blocks people).

P/E/S/T	Political, Economic, Social, Technological (as PEST)
Environmental	Sustainability, climate, environmental regulation
Legal	Employment law, consumer protection, IP law

System Thinking: your user lives in a network — you also "sell" to their parents, boss, culture. **If the user says YES but their environment says NO, the idea is dead.**

L3 · Jobs-to-be-Done & Kano P3

JTBD: people "hire" a product to do a job — **Functional** (practical task), **Social** (how they appear), **Emotional** (how they feel).

Not a drill — a hole to hang a photo to feel like a good parent.

Required	Must-have; absence = dealbreaker
Expected	Baseline; assumed (anger if absent)
Desired	Nice-to-have; preferred
Unexpected	Delighters; didn't know they wanted it

Kano drift: today's **delighter** → tomorrow's **expected** → eventually **required**.

L3 · ★ Problem Statement Formula P3

[User] struggles with [situation] because [root cause], leading to [consequence].

Good vs Bad

Good: "Busy students skip meals during peak hours because food queues are unpredictable, leading to fatigue and reduced academic performance."

Bad: "Students need better campus food." → vague, no root cause/consequence.

Specific user + situation + root cause + consequence. Vague statements → irrelevant solutions.

L4 · Personas & the Beachhead P3

Myth of "everyone": if your customer is everyone, you have no customer. "Students/People/Users" = **categories, not segments**.

A real user is in a specific situation, doing something specific, with a real struggle. "*If you can't see the moment, you don't understand the problem.*"

Persona: When (situation) · What they do (behaviour) · What they avoid (pains) · What they want to feel (emotional job).

Beachhead persona = Segment + real human + real moment + job + intensity.

Shark bite (urgent, act NOW) vs **Mosquito bite** (annoying, ignored). Beachhead needs a shark-bite problem.

L4 · Root Cause & Jargon Assassin P3 P9

5 Whys: ask "why?" repeatedly to drill from symptom to **root cause**; then move from "**What is**" (problem) to "**What if**" (solution).

Jargon Assassin: jargon is a **symptom**; unclear thinking is the real problem. If you can't explain it to a 10-year-old or your grandmother in Amman, the idea is too weak. **"Speak Human, Not Corporate."**

L5 · ★ How Might We (HMW) — 4 Types P4

HMW = a **decision about which part of the problem space to explore**; focuses thinking while keeping directions open. *"How might we help [user] achieve [job] despite [pain]?"*

1 — Efficiency	"reduce waiting time?" → faster/optimise
2 — Experience	"reduce stress while waiting?" → emotional
3 — Behaviour	"change when students eat?" → shift habits
4 — System	"redesign food distribution?" → structural

⚡ The HMW **type you choose determines the kind of solutions** you'll get.

L5 · Divergence, Ideation & Communication P4 P9

Divergence = expand thinking. The first idea is obvious — push for **many fundamentally different** ideas. Rules: quantity over quality, no judging, different directions.

Wrong (same thinking): app with notifications, app with tracking, app with AI. **Right** (different thinking): eliminate queues, shift eating times, remove the cafeteria need.

Without HMW: different but irrelevant. With HMW: different **and** relevant.

Communicating: fillers (umm, like) = panic; **the Pause** = confidence. Slow down at the insight (the "because").

L6 · Validation Engine & Double Diamond P4 M2 D2

Insight → Hypothesis → Prototype → Test → Learn → Decide

Discover	Expand — behaviour, tension, contradiction
Define	Focus — construct the right problem (**hardest step**)
Develop	Expand — the HMW zone
Deliver	Focus — test, refine, decide

⚡ Fatal mistake: testing **HOW** (usability) before **WHY** (value). Confirm they want it, then refine how. "If I removed the tech, would it still work?"

L6 · Prototype Types & Demand Signals P4 M2

Type	Tests · limitation
Storyboard	Journey/logic (6–8 steps + emotion) · can't test demand
Role-Play	Service/conversation · depends on acting
Interface (paper/digital)	Screens/flow — wrong taps, hesitation · not real speed
Digital MVP	Real interaction: 1 screen/flow/function · risk of overbuilding
Landing / Fake Door	Demand — "sign up", watch clicks · tests intent not behaviour

Signals: "I would use this" → **ignore** (cheap talk); "Interesting, but..." → **dig**; "When can I get it?" → **act** (real demand). Patterns > opinions.

L7 · Feedback Extraction & Evidence Chain M2 D2

Feedback = **extraction** of behaviour, friction, expectation gaps, commitment. You tested **assumptions**, not ideas. Testing without changing anything = performance.

Observation → Insight → Decision → Alignment

Observation (what they did) · **Insight** (what it means) · **Decision** (what you changed) · **Alignment** (why it fits better). Skip one = guessing.

"If you can't say this, you didn't learn": "We observed ___ · This means ___ · So we changed ___ · This aligns with the user because ___."

L7 · Feedback Methods & RAT M2 D2

Silent Observation	No help/guidance — watch hesitation, wrong actions, breakdown.
Thinking Aloud	User narrates → where their mental model breaks.
Wizard of Oz	Simulate the experience manually → does the value hold without tech?
Commitment Test	Push for action/effort; refusal invalidates earlier praise.

RAT — Riskiest Assumption Test: the one assumption where "if THIS is wrong, everything collapses" — test it first. Set success rate & failure threshold *before* testing; test only your persona; never ask "Do you like it?"

L7 · ★ Iterate / Pivot / Kill D2

Signal	Diagnosis → Action
Usability Friction (HOW)	Wants value but blocked → **Iterate** (fix flow)
Value Deficit (WHY)	Understands but doesn't care → **Pivot** (weak problem)
Mental Model Mismatch	Uses it differently → **Investigate** (new opportunity)

Decision Layer: Iterate = problem real, solution weak · Pivot = problem wrong · Kill = no behavioural intent after multiple tests.

Cost of being wrong: MVP (small, iterate easily) → Pilot (controlled) → Open Beta (everyone sees) → Polished Launch (can't fix fast = danger). *"Launch when you can't iterate = gambling."*

⚡ EXAM STRATEGY — how to attack each question type + the traps that lose marks

🎯 How to answer each question type

- **Classify (SWOT/4Ps/BMC):** internal-vs-external & good-vs-bad (SWOT); which lever changed (4Ps); match keyword to definition.
- **Which pricing strategy:** low to win share = Penetration; high then drop = Skimming; cost-based = Cost-Plus; value-based = Value-Based.
- **Calculation:** write the formula first, plug numbers, show working. Breakeven, margins, ROI, TAM/SAM/SOM all have fixed formulas.
- **Best single BMC change:** the option that fixes desirability + feasibility + viability together.
- **Design "what should they conclude":** translate the behaviour/complaint into the underlying **need**; never take the wish literally.
- **Iterate/Pivot/Kill:** HOW-problem = Iterate; WHY-problem = Pivot; repeated no-intent = Kill.

⚠️ Most-confused distinctions

PEST vs PESTEL	PESTEL = PEST + Environmental + Legal
Fixed vs Variable	Fixed = same regardless of volume; Variable = scales with units
CapEx vs OpEx	CapEx = long-term asset (oven); OpEx = day-to-day cost
Penetration vs Skimming	Low to grab share vs high first then lower
Pain vs Gain	Pain = avoid; Gain = desired positive outcome
Strength vs Opportunity	Internal + vs external +
Weakness vs Threat	Internal − vs external −
Profit vs Cash Flow	Can be profitable yet cash-broke (timing)
TAM/SAM/SOM	Whole vs serviceable vs obtainable slice
Iterate vs Pivot	Weak solution (fix HOW) vs wrong problem (change WHY)

⚠️ Calculation traps that lose marks

- **Breakeven:** compare **total contribution** (CM×units) to fixed costs — not revenue.
- **Reclassification:** moving a cost COGS↔OpEx **never changes Net Profit** — only GPM moves.
- **Margins:** always divide by **revenue**; same profit on lower revenue = higher margin.
- **ROI vs NPM:** different denominators — test each threshold separately.
- **Price ↑ in TAM/SAM/SOM:** "willing & able" % falls → SAM & SOM shrink.
- **"At least 12%":** exactly 12% **meets** it (no shortfall, no headroom).
- Convert monthly→annual revenue by ×12 when asked for annual SAM revenue.

🔑 High-yield one-liners

- Innovation must create **value**, not just be invented.
- Good innovation = Desirable ∩ Feasible ∩ Viable.
- No defined segment = **no customer**.
- Observed behaviour **>** stated preference.
- Test **WHY (value)** before **HOW (usability)**.
- "I would use this" = cheap talk; "When can I get it?" = real demand.
- Beachhead needs a **shark-bite** (urgent) problem.
- **Define** is the hardest step — construct the right problem.
- If user says YES but their system says NO → the idea dies.
- Lean = validate fast; Agile = adapt in cycles; they complement.
- Profitable ≠ cash-positive (timing of cash flows).